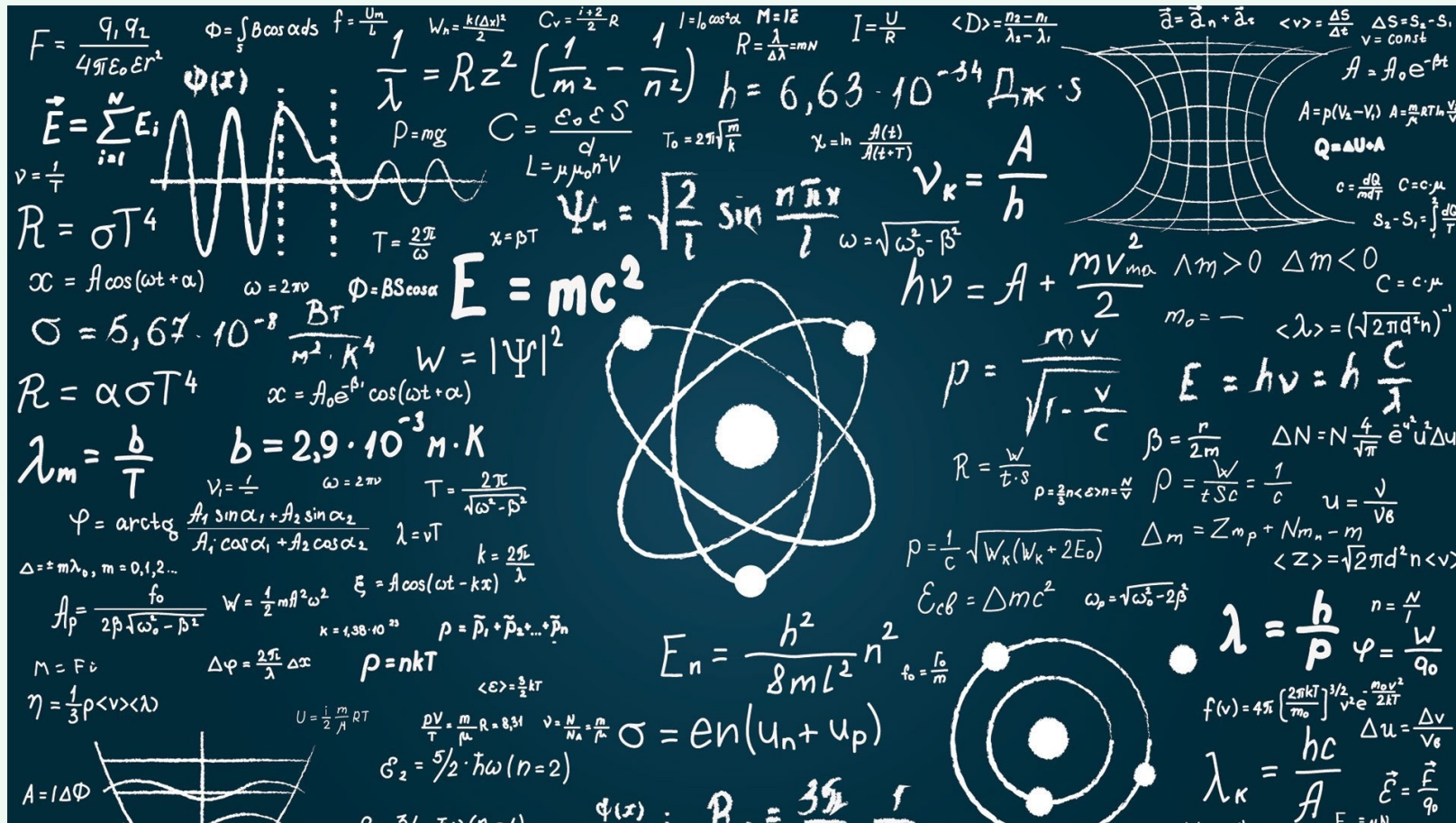


Quantum Mechanics 2



Lecture 3
(Autumn Term, 2025)

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Recap of Lecture 2

In QM, form of wave eqn. is a basic postulate, cannot be derived from other equations. Following a plausibility argument and satisfying constraints (energy conservation, DeBroglie relations, linearity), we get time-dependent Schrodinger's equation:

$$\frac{-\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V(x,t)\psi(x,t) = i\hbar \frac{\partial \psi(x,t)}{\partial t}$$

S.E. is a complex 2-order partial differential equation.
Its solution is a complex function.

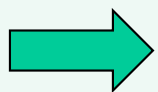
SE solution for a free particle ($V=V_0$) is $\psi(x,t) = Ae^{i(kx-\omega t)}$

giving $E_{tot} = \hbar\omega = \frac{\hbar^2 k^2}{2m} + V_0 = \frac{p^2}{2m} + V_0$

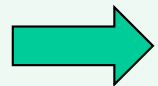
QM postulate 1: wavefunction is a complete description of the state of the quantum system, containing all the physical information it is possible to know about the object.

However, any physical observable must be real so the wavefunction itself is not an observable, i.e. cannot be measured (unlike a classical wave amplitude).

Must apply mathematical operations on wavefunction to extract physical information from it.



Probability density function (pdf)



Operators and Expectation values

Define:

Probability Density Function (pdf) = $P(x,t) dx$

as the probability of measuring particle between x and $x + dx$

$P(x,t)$ must be real, ≥ 0 and normalised to 1, i.e.

$$\int_{-\infty}^{+\infty} P(x, t) dx = 1$$

since the probability of finding particle must be unit

In QM we define: $P(x, t) = |\psi|^2 = \psi^* \psi$

c.f. classically, intensity is proportional to (amplitude)²

Expectation value:

Suppose we make many measurements on particles in the same state.

We can predict the mean result of measurements over many experiments, called **expectation value**.

Wavefunction gives probability of result of measurement, but cannot give definite result for a single measurement.

We can use pdf to find expectation value for a certain quantity we want to measure

QM Postulate 2:

For every physical observable, there is an associated mathematical operator.

When ψ is an eigenfunction of a particular observable, the result of the operator acting on ψ is to return the value of the observable multiplied by ψ

Summary

QM postulate 1: wavefunction is a complete description of the state of the quantum system, containing all the physical information it is possible to know about the object.

QM Postulate 2: For every physical observable, there is an associated mathematical operator.

We defined Probability Density Function $P(x, t) = |\psi|^2 = \psi^* \psi$

and Expectation Value $\langle f(x) \rangle = \int_{-\infty}^{\infty} \psi^* f(x) \psi dx$

$$\langle \hat{O} \rangle = \int_{-\infty}^{\infty} \psi^* \hat{O} \psi dx$$

and Operators:

Momentum $\hat{p} = -i\hbar \frac{\partial}{\partial x}$

Kinetic Energy $\frac{\hat{p}^2}{2m} = \frac{-\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$

Total energy $\hat{E} = i\hbar \frac{\partial}{\partial t}$

Next time:
eigenfunction
& eigenvalue